

Linear Systems

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Learning Outcomes

At the end of this lecture, you should be able to:

- understand what is a system of linear equations and its solutions.
- Write the augmented matrix of a system of linear equations and vice versa.

Introduction

In high school, you have learned to solve some two-by-two system of linear equations such as

$$\begin{aligned} 2x + y &= 4, & \text{--- ①} & \quad 4x + 2y = 8 \\ 4x - 2y &= 0. & \text{--- ②} & \end{aligned}$$

There were two ways of solving this:

Substitution:

From ①, $y = 4 - 2x$. Substitute this into ②,

we get $4x - 2(4 - 2x) = 0 \Rightarrow 4x - 8 + 4x = 0 \Rightarrow x = 1$

So, $y = 4 - 2(1) = 2$

Elimination:

Take $2 \times \text{①} - \text{②}$, we get $4y = 8 \Rightarrow y = 2$

Take $2 \times \text{①} + \text{②}$, we get $8x = 8 \Rightarrow x = 1$.

Introduction

This is easy to be done when we have a system with small amounts of equations with small amount of unknown.

What about when we have large amount of equations and unknowns, says 20 equations with 20 unknowns?

Even this would be “hard”

$$\begin{aligned}w + 2x + 3y + 4z &= 1, \\ -x + 5y + 6z &= 2, \\ y + 7z &= 3.\end{aligned}$$

In this chapter, we will learn some systematic way to solve the systems of linear equations.

Linear Equation

A **linear equation** in n variables $x_1, x_2, \dots, x_n$ has the form

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b$$

where $a_1, a_2, \dots, a_n$ and b are real constants.

An a_i , $1 \leq i \leq n$, is called a **coefficient**, and the n variables in the linear equation are called the **unknowns**.

Example:

1. The equations $4x - 2y = 1$, $2x + 3y - 5z = 7$, and $x_1 - x_2 + 2.5x_3 + x_4 = 10$ are linear equations.
2. The equations $xy = 1$, $x^2 + y^2 = 9$, $\sin x + \cos y = 1$ and $x - e^y = 0$ are non-linear.

Solution of a Linear Equation

Given n real numbers $s_1, s_2, \dots, s_n$. We say that

$x_1 = s_1, x_2 = s_2, \dots, x_n = s_n$ is a **solution** of a linear equation

$a_1x_1 + a_2x_2 + \dots + a_nx_n = b$ if the equation is satisfied when we substitute the values into the equation accordingly.

$$\text{Eg: } 2x + y = 4 \quad \left. \begin{array}{l} x=0, y=4 \\ x=1, y=2 \end{array} \right\} \text{solution}$$

$x=0, y=2 \leftarrow \text{not a solution.}$

The set of all solutions of the equations is called the **solution set** of the equation and an expression that gives us all these solution is called the **general solution** of the equation.

$$\text{Eg: } 2x + y = 4$$

$$\text{solution set: } \left\{ \begin{pmatrix} x \\ 4-2x \end{pmatrix} \right\} \text{ or } \{ (x, y), x=t, y=4-2t \}$$

Solution of a Linear Equation

Example:

Consider the linear equation $4x - 2y = 1$.

$x=0, y=-\frac{1}{2}$; $x=\frac{1}{4}, y=0$; and $x=1, y=\frac{3}{2}$ are solutions of $4x-2y=1$.

In fact this linear equation has infinitely many solutions.

$x=t$ and $y=2t-\frac{1}{2}$ where $t \in \mathbb{R}$ is an arbitrary parameter is called the general solution of the linear equation $4x-2y=1$ and $\{(x,y) \mid x=t, y=2t-\frac{1}{2}, t \in \mathbb{R}\}$ or $\left\{\begin{pmatrix} t \\ 2t-\frac{1}{2} \end{pmatrix}, t \in \mathbb{R}\right\}$ is the solution set of the linear equation $4x-2y=1$.

① $x = \frac{1+2s}{4}$ $y=s, s \in \mathbb{R}$.

System of Linear Equations

A finite set of m linear equations in n unknowns $x_1, x_2, \dots, x_n$ is called a **system of linear equations** (or **linear system**):

$$a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2$$

$$\vdots$$

$$a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m$$

where a_{ij} and b_i , $1 \leq i \leq m$, $1 \leq j \leq n$ are real constants. An a_{ij} is called a **coefficient**, b_i is called a **constant** and x_j is called an **unknown** (or **variable**).

Given n real numbers $s_1, s_2, \dots, s_n$. $x_1 = s_1, x_2 = s_2, \dots, x_n = s_n$ is a **solution of the system** if it is a solution of every equation in the system.

The set of all solutions of the system is called the **solution set** of the system and an expression that gives us all these solution is called the **general solution** of the system.

System of Linear Equations

Example:

Consider the system of linear equations: $x - 4y + 7z = 5$,
 $2x + 15z = 10$.

$x = 5, y = 0, z = 0$ is a solution of the system because it satisfies both equations, but $x = 9, y = 1, z = 0$ is not a solution of the system because it does not satisfy $2x + 15z = 10$.

Not all system of linear equations have solutions. For example: the system of linear equations

$$\begin{aligned}x - 2y + 3z &= 1 \\ -2x + 4y - 6z &= 5\end{aligned}$$

has no solution.

A system of linear equations that has at least one solution is called **consistent**. A system that has no solution is called **inconsistent**.

Every system of linear equations has either no solution, exactly one solution or infinitely many solutions.

Augmented Matrix

A system of m linear equations in n unknowns $x_1, x_2, \dots, x_n$:

$$a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2$$

$$\vdots$$

$$a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m$$

is equivalent to the matrix equation $AX = B$ where

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix} \text{ is the coefficient matrix, and}$$

$$X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \text{ and } B = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix} \text{ are the column vectors of the unknowns and of}$$

the constants, respectively.

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1$$

$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = b_2$$

$$a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3$$

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}, \quad X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \quad b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

$$AX = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$= \begin{bmatrix} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = b$$

Augmented Matrix

If $B = 0$ ($b_i = 0$ for all i), the system is said to be **homogeneous**.

If $B \neq 0$ (there is a $b_i \neq 0$), the system is said to be **non-homogeneous**.

This system of m linear equations can be represented by a rectangular array numbers

$$[A \mid B] = \left[\begin{array}{ccccc|c} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} & b_m \end{array} \right]$$

This array is called the **augmented matrix** of the system.

Example:

Write the augmented matrix for the system of linear equations

$$-x_1 + x_2 + 2x_3 = 2,$$

$$3x_1 - x_2 + x_3 = 6,$$

$$-x_1 + 3x_2 + 4x_3 = 4.$$

$$\left[\begin{array}{ccc|c} -1 & 1 & 2 & 2 \\ 3 & -1 & 1 & 6 \\ -1 & 3 & 4 & 4 \end{array} \right]$$